

Research strategy in an AI-capable world

The recommendation is to build a research program on reliable learning and inference from imperfect, adaptive information. For Cong Ma, the best opportunities sit where three existing strengths meet: sharp statistical limits, algorithms that exploit structure, and learning from data that were selected or generated by another system. AI makes this intersection more consequential because models increasingly supply the representations, demonstrations, evaluations, and decisions on which later learning depends.

The immediate priorities should be: reliable evaluation of changing AI systems; learning from demonstrations and imperfect verification; and multimodal transfer that preserves information needed for downstream decisions. Adaptive allocation of computation and human feedback should connect the first two. Formal verification should become part of the group's research practice, with a small, selective investment in reusable statistical proof infrastructure.

This is a continuation of an existing trajectory. Early work explained why simple nonconvex algorithms succeed; subsequent work examined when data from a different distribution or policy are useful; recent work asks which information should be retained and which learning objectives are appropriate. The common intellectual contribution is to identify the condition that changes what is statistically possible or which procedure is justified. That is a stronger organizing principle than any single application label.¹

The main adjustment is in project selection. Give priority to results that change a learning objective, identify an unavoidable information requirement, or support a consequential decision. Proof difficulty alone is a less secure reason to invest several years. AI can accelerate useful technical work while also increasing the supply of technically competent papers; the relative value of importance, sound formulation, and independent evidence is likely to rise. This is a strategic judgment, not a forecast that any part of research will remain exclusively human.

This assessment considers a three-to-five-year horizon, using evidence available on 14 September 2026. The trajectory review covers all 47 entries in the supplied publication inventory, representative paper abstracts, and selected full-text formulations and discussions. It is a strategic synthesis, not a proof audit or an attribution of individual contributions within coauthored work. Proposed projects are hypotheses to investigate; their novelty still requires a focused review at the theorem level.

Priority	Recommended role	Why it fits
Reliable AI evaluation	First flagship	Off-policy inference, ranking, testability, robustness
Demonstrations and verification	Second flagship	Reward-class learning, pessimism, distribution shift
Multimodal transfer	Focused continuing program	Shared subspaces, intrinsic dimension, misspecification
Compute allocation; verified statistics	Connecting projects and research practice	Batched decisions; existing Lean experience

1. Cong Ma, *Publication inventory and research profile*

Your trajectory has a clear statistical center

2015-2020: computation, geometry, and sharp estimation. The inventory is dominated by low-rank estimation and related optimization. The implicit-regularization work supplies a way to analyze algorithm trajectories under statistical dependence. Matrix-completion inference moves from estimating a matrix to quantifying what can be known about its entries. Scaled gradient descent changes the computational procedure to address ill-conditioning. Together these establish expertise in connecting an algorithm's actual behavior to statistical precision.²⁻⁴

2021-2023: information quality becomes central. Offline RL, off-policy evaluation, and covariate shift broaden the focus from a single sampling model to mismatches between available data and the target task. The pessimism work connects nearly expert data to broader offline learning. The RKHS and parametric covariate-shift results show that whether reweighting is necessary depends on the model and its specification. This is a useful research habit: determine when a popular correction is needed and when a simpler estimator already suffices.⁵⁻⁷

2024-2026: representations, evaluation, and the meaning of successful learning. Adaptive off-policy estimation links inference to online learning. The demonstration paper uses reward-class complexity to formalize successful answering without requiring imitation of the demonstrator's entire distribution. The multimodal papers develop complementary accounts of shared information, modality-specific information, and noisy subspace intersection. These threads increasingly concern how one learned system supplies useful information to another.⁸⁻¹³

The counts below are descriptive. They use the repository's year fields, which often precede journal publication, and its single topic label per entry. They include a survey and commentaries. They should not be interpreted as impact scores, a complete account of ongoing work, or evidence that one period was more productive than another.

The comparative advantage suggested by this record is mechanism-driven statistical theory. Several papers reveal that a seemingly natural procedure can fail for a precise reason, then identify an alternative or a boundary of possibility. A promising next step is to apply that habit to systems that generate their own training signals and use their own predictions to determine what evidence is collected next. This joins the existing fields without requiring a wholesale change of research identity.

Inventory period	Records	Largest topic groups
2015-2020	16	Low-rank estimation 9; scaled gradient descent 2
2021-2023	16	RL/bandits 6; transfer 3; scaled gradient descent 3
2024-2026	15	Multimodal 4; RL/bandits 4; transfer 3

2. Ma, Wang, Chi and Chen, *Implicit Regularization in Nonconvex...*
3. Chen, Fan, Ma and Yan, *Inference and Uncertainty Quantification...*
4. Tong, Ma and Chi, *Accelerating Ill-Conditioned Low-Rank...*
5. Ma et al., *Optimally Tackling Covariate Shift in...*
6. Ge et al., *Maximum Likelihood Estimation is All You...*
7. Rashidinejad et al., *Bridging Offline Reinforcement Learning...*

8. Lee and Ma, *Off-Policy Estimation with Adaptively...*
9. Joshi et al., *Learning to Answer from Correct...*
10. Gui, Ma and Ma, *Multi-Modal Contrastive Learning Adapts...*
11. Gui, Ma and Ma, *IndiSeek Learns Information-Guided...*
12. Yang and Ma, *Estimating Shared Subspace with AJIVE:...*
13. Song, Yang and Ma, *Bias-Corrected Subspace Intersection:...*

The results that offer the strongest next questions

Covariate shift: the right correction depends on specification. The RKHS paper identifies regimes where regularization or truncated reweighting is needed for optimal performance. The parametric paper shows that source-only maximum likelihood can already be optimal under correct specification, while misspecification changes the answer. The strategic lesson is to analyze how much target information is necessary to decide whether a pretrained model needs correction. A general robustness penalty is less informative than a justified distinction between these regimes.^{5,6}

BCSI: a small-looking perturbation can change the optimal rate. The September 2026 paper identifies second-order leakage bias in plug-in subspace intersection and a second-order minimax term associated with nearly aligned view-specific subspaces. Its bounds allow heterogeneous view dimensions and signal strengths, with rate matching under the stated rank comparability condition. The future question is whether these geometric effects predict the downstream value of an additional view, including when only one view carries a weak target signal.¹³

Trade-off testing: separate validity from what can be detected. The 2026 paper identifies finite VC dimension as the testability threshold within its exact-attainability framework. Type I error control does not require attainability, but power does. This is a valuable template for AI evaluation: distinguish a valid procedure from one capable of detecting the failures that matter. The application is specific binary-testing theory; broader claims about certifying arbitrary AI behavior require new assumptions and analysis.¹⁵

Batched bandits: feedback opportunities have statistical value. The unknown-margin work finds a polynomial adaptation penalty in the studied batched setting and characterizes it through a variational problem. The barrier disappears, up to polylogarithmic factors, with sufficiently many batches of order $\log \log T$. This suggests studying whether additional feedback rounds in reasoning or auditing are fundamentally different from spending the same budget on more parallel attempts. The analogy identifies a research question rather than transferring the theorem to language models.¹⁹

Lasso: established methods still deserve adversarial scrutiny. A polynomial rate gap for the Lasso in the studied design regime, despite data-dependent tuning, is a strong example of classical theory with a clear consequence. Similar investigations of ubiquitous AI procedures could ask whether the limitation comes from tuning, objective choice, or unavailable information. The resulting contribution should distinguish these explanations and show what a better procedure must change.¹⁸

Taken together, these results suggest a project-selection principle: favor a surprising separation that can be explained simply and that changes a decision. This is a pattern in the reviewed work, not a claim that these papers are its most influential contributions or that every next paper should follow the same form.

5. Ma et al., *Optimally Tackling Covariate Shift in...*

6. Ge et al., *Maximum Likelihood Estimation is All You...*

13. Song, Yang and Ma, *Bias-Corrected Subspace Intersection:...*

15. Shi, Wang and Ma, *When Are Trade-Off Functions Testable...*

19. Jiang and Ma, *The Adaptivity Barrier in Batched...*

18. Pathak and Ma, *On the Design-Dependent Suboptimality of...*

AI changes the value of theoretical work

Research-level proof production is already affected. First Proof's June 2026 evaluation used ten solved but unpublished research problems, with organizers running the systems and experts grading their solutions. Across the four systems, seven problems received at least one passing submission; passing included solutions requiring minor revisions. The report also describes serious failures. This establishes capability on a restricted collection of research proof tasks, not autonomous success at choosing important problems or running a complete research program.²²

The August 2026 study by Jin, Ke, and Sui identified 1,712 mathematics submissions with disclosed substantive AI contributions among 32,944 arXiv submissions in its sampling window. These are disclosure-based classifications, not independently verified contributions. The evidence supports taking adoption seriously while leaving large uncertainty about correctness, selection effects, and the amount of human intervention.²³

Your own work supplies the most relevant evidence. The BCSI paper explicitly discloses AI assistance with proof writing and formalization. Its public repository describes checked upper- and lower-bound entry points and reports only standard Lean axiom dependencies. This assessment has not rebuilt those targets or independently checked that every formal statement matches the manuscript. The practical inference is nevertheless clear: AI-assisted theoretical statistics is already part of your research, rather than a distant possibility.^{13,14}

Expect the return to some tasks to change. Routine derivations, coding, literature triage, familiar proof templates, and initial simulations are natural candidates for substantial acceleration. The size of that acceleration will vary by task and supervision. Deep understanding remains useful because it helps detect invalid assumptions, formulate counterexamples, and decide whether an argument actually resolves the scientific question. These are complementary skills, not a guarantee against future automation.

A distinction matters: verifying a formal theorem establishes the theorem relative to its definitions and assumptions. It does not establish that a representation models the application correctly, that a statistical target answers the important question, or that the result improves a real decision. AI may help with all these activities, but evidence from proof benchmarks does not settle them. A good program should therefore connect formal validity with empirical and scientific relevance.

Plan for several futures. If progress slows, the proposed problems remain standard, substantial statistical questions. If proof and experiment automation becomes much stronger, the same program can examine more models, more counterexamples, and more demanding guarantees. If research becomes broadly automated, there is no defensible assurance that a particular theory niche is protected; close scientific collaborations, independent evaluation, and ownership of useful research infrastructure provide options without depending on that assurance.

22. Abouzaid et al., *First Proof Second Batch*

23. Jin, Ke and Sui, *The Gold Rush in AI4Math: Where Are We Now?*

13. Song, Yang and Ma, *Bias-Corrected Subspace Intersection:...*

14. Cong Ma et al., *BCSI Lean Verification*

Where to continue, and where to narrow the scope

Keep the technical foundations; make each field earn its place through a consequential question. The portfolio is already broad enough. The next gain is more likely to come from selecting projects that combine existing strengths than from adding another general field. In particular, transfer learning can become a property studied across the program: when does external information help, when does it harm, and can that distinction be learned from limited target evidence?

Low-rank theory should remain an active source of methods when it exposes a new phenomenon. The BCSI bias correction and inexact-side-information work provide current examples of questions about information quality and interaction between views. An extension whose only contribution is a slightly sharper bound in an otherwise settled model deserves a higher bar for priority. The distinction is the consequence of the result, not whether its title contains AI.^{13,16}

Your Lasso result illustrates why classical statistics can still be a strong investment: it identifies a polynomial suboptimality of a ubiquitous procedure for a specific design regime. Similar surprises can matter more than a theorem about a fashionable architecture. Preserve room for such work when the counterexample or principle changes how the field should think.¹⁸

Ranking and item-response theory are unusually useful connections to AI measurement. However, model ability should not automatically be treated as a single stable scalar. Task family, inference budget, model updates, and judge behavior can alter comparisons. The compelling extension is to characterize when a cheaper evaluation preserves a decision under these changes, using ranking and inference tools where their assumptions can be defended.^{17,30,31}

A focused application partnership would improve this portfolio. Multimodal biological measurement is a plausible option because IndiSeek already includes CITE-seq experiments; another is AI evaluation with access to repeated model and evaluator updates. Choose one application where a collaborator can define the scientific target and obtain credible validation data. This is a recommendation about access and feedback, not an inference that such a collaboration is currently available.¹¹

Current field	Recommended direction
RL and bandits	Continue: feedback quality, adaptive evidence collection, resource allocation
Transfer and robustness	Deepen: target information requirements and adaptation under misspecification
Multimodal learning	Continue: downstream value of shared and modality-specific information
Low-rank estimation / optimization	Selective: new statistical mechanisms and consequential algorithms
Ranking / uncertainty quantification	Connect: reliable, economical evaluation of changing systems
Formal verification	Integrate across projects; keep infrastructure work bounded

13. Song, Yang and Ma, *Bias-Corrected Subspace Intersection*:...

16. Yang and Ma, *Sample-Efficient Inductive Matrix*...

18. Pathak and Ma, *On the Design-Dependent Suboptimality of*...

17. Yang and Ma, *Random Pairing MLE for Estimation of*...

30. Maia Polo et al., *tinyBenchmarks: Evaluating LLMs with*...

31. Li et al., *Adaptive Testing for LLM Evaluation: A*...

11. Gui, Ma and Ma, *IndiSeek Learns Information-Guided*...

First flagship: reliable evaluation of adaptive AI systems

The question is how to certify an improvement when the model, evaluator, and evaluated tasks are all being selected using earlier results. A concrete setting is a sequence of model or prompt updates, cheap scores from one or more AI judges, and a limited budget of trusted outcome checks. The target must be explicit: for example, the improvement of a proposed update on a fixed deployment population, or the probability that a deployment rule accepts a worse model.

This direction combines adaptive off-policy estimation, uncertainty quantification, distributional robustness, and ranking. The trade-off-function paper contributes a particularly useful style of question: identify which structural assumptions make a guarantee testable at all. Its current results concern binary-testing frontiers, including privacy auditing; applying that reasoning to AI reliability is a new problem, not a direct corollary.^{8,15,17,20,21}

The entry point must be narrower than adaptive AI evaluation. Prediction-powered inference already uses imperfect predictions with trusted labels. Active inference, robust sampling, R-AutoEval+, and multiwave inference already address important combinations of selective labeling, adaptivity, and efficiency. tinyBenchmarks and ATLAS already use economical evaluation and item-response ideas. Simply combining a judge with a confidence interval, or choosing difficult test items, would have limited distinctiveness.^{24-27,29-31}

A promising first project is *valid model comparison with adaptively updated judges and batched expert audits*. Start with bounded scores and a fixed target population. Each proposed model is chosen using past information; its deployment claim is evaluated using fresh or appropriately protected evidence. Judge predictions may change between rounds. Trusted checks are sampled with known, positive probabilities that can depend on the available history. The research goal is a finite-sample guarantee for decisions over time, paired with an audit-allocation rule and a lower bound for the cost of evaluator drift or limited update opportunities.

The first theorem should separate bias correction from the price of selecting models and allocating audits. Begin with the simplest feedback structure in which a naive method fails, then establish what additional assumption or fresh evidence repairs it. Do not assume a familiar confidence sequence automatically handles reuse of outcomes by an adaptively chosen target model. Likewise, without any reliable anchors or restrictions on judge error, agreement among judges cannot identify true quality.

Evidence and stopping rule. Compare against random expert audits, PPI-style correction, robust active sampling, and R-AutoEval+ under matched total costs. Measure wrong deployment decisions, interval coverage, audit burden, and sensitivity to common judge errors. Use repeated simulations for statistical validity and at least one application with independently checked outcomes. Continue only if the joint adaptive setting creates a meaningful unresolved guarantee or a material saving; if an existing method already covers it, narrow the claim or redirect the project.

8. Lee and Ma, *Off-Policy Estimation with Adaptively...*

15. Shi, Wang and Ma, *When Are Trade-Off Functions Testable...*

17. Yang and Ma, *Random Pairing MLE for Estimation of...*

20. Gui, Barber and Ma, *Conformalized Matrix Completion*

21. Gui, Barber and Ma, *Distributionally Robust Risk Evaluation...*

24. Angelopoulos et al., *Prediction-Powered Inference*

25. Zrnic and Candes, *Active Statistical Inference*

26. Li, Zrnic and Candes, *Robust Sampling for Active Statistical...*

27. Park et al., *Adaptive Prediction-Powered AutoEval...*

29. Kluger and Bates, *M-Estimation under Two-Phase Multiwave...*

30. Maia Polo et al., *tinyBenchmarks: Evaluating LLMs with...*

31. Li et al., *Adaptive Testing for LLM Evaluation: A...*

Second flagship: learning from demonstrations and imperfect verification

The question is which information makes successful answering learnable, and which information requires interaction. Your demonstration paper distinguishes correctness complexity from the complexity of demonstrator behavior. Its limitations are important: naive implementation scales with reward-class cardinality, and efficient implementations for practical classes remain open. Its pass@k extension already covers that metric; changing the metric alone is not a new direction.⁹

The highest-value extension would connect this formulation to a realizable post-training procedure. Begin with a manageable reward class, imperfect demonstrations, and limited access to trusted verification. Seek an algorithm whose sample and computational costs depend on meaningful reward structure and on the availability of useful candidate answers. A tractable result for linear or low-rank rewards could be worthwhile if it predicts behavior under controlled departures from those assumptions.

Distinguish error that averages out from error that optimization exploits. Plesner and colleagues find that substantial verifier noise can be tolerated in their code and scientific-reasoning experiments. Helff and colleagues exhibit systematic verifier shortcuts on inductive-reasoning tasks. These results are compatible: randomly dispersed mistakes and errors concentrated on attractive policy outputs can have very different consequences. Neither supports a universal conclusion about the safety or futility of imperfect verification.^{32,33}

A concrete first project is *learning under policy-dependent verifier error with a limited trusted-feedback budget*. Compare two verifiers with the same accuracy on an initial distribution but different errors on candidates reached during optimization. Characterize the conditions under which an algorithm can improve true success, and the conditions under which it must request additional information. Separate misspecification of the reward class, failure to generate useful candidates, verifier error after selection, and optimization error. A matching impossibility construction would identify which limitation extra computation cannot remove.

The experimental design should vary demonstration diversity, task shift, verification precision, and the frequency of independent checks. Compare likelihood-based supervised fine-tuning, filtering and retraining, an established RL method, and the proposed tractable learner. Report both single-attempt success and success under a fixed inference budget, alongside selection error from the verifier. Include held-out task families and checks the learner cannot optimize against. A gain at one sampling budget does not by itself establish new reasoning capability.

This last distinction is supported by conflicting empirical findings: one study finds that large-budget sampling of base models can recover or exceed the tested RL models' coverage, while exploration-oriented work reports improvements beyond the tested base performance. The studies do not prove a universal capability boundary. They motivate precise claims about reachable candidates, budget, and feedback rather than an unqualified claim that RL can or cannot create reasoning.^{34,35}

Continue if the work yields a tractable separation or an actionable rule for allocating trusted feedback. If a proposed noise model is merely independent label noise with established guarantees, seek the adaptive failure mechanism before extending the proof.

9. Joshi et al., *Learning to Answer from Correct...*

32. Plesner et al., *An Imperfect Verifier is Good Enough:...*

33. Helff et al., *LLMs Gaming Verifiers: RLVR can Lead to...*

34. Yue et al., *Does Reinforcement Learning Really...*

35. Deng, Wei, Yu and Wu, *Unlocking Reasoning Capabilities in LLMs...*

Third priority: multimodal transfer that preserves task-relevant information

The question is when an additional modality or learned representation improves a downstream decision. Your portfolio now connects shared-subspace estimation, intrinsic dimension in contrastive learning, and extraction of modality-specific information. That combination supports a distinctive program on the statistical value of heterogeneous views. It is particularly useful when a powerful pretrained representation is available but only limited target labels can establish whether its information is appropriate.^{10-13,16}

A crucial distinction is between information shared by modalities and information useful for a task. Shared variation can include nuisance effects; a rare target signal may appear in just one modality. A useful joint signal can also require both modalities. The latter phenomenon is already treated in work such as CoMM, so demonstrating that shared-only representations can lose useful information is a starting example, not by itself a sufficiently new contribution.^{11,36}

A concrete first project is *adaptive transfer with imperfect shared and modality-specific representations*. Begin with two noisy views, a small labeled target sample, and controlled discrepancies between source and target. Let the target depend on both common and view-specific directions. Compare a target-only estimator, a shared-only estimator, simple concatenation, and a method that chooses how strongly to use each source of information. The goal is to identify when the target labels are sufficient to learn which transfer strategy is useful.

A meaningful theorem would compare downstream risk to an oracle choice among these strategies, with an explicit price for adaptation and misspecification. It should identify regimes where adding a weak view helps, does essentially nothing, or makes an inadequately regularized procedure worse. Selection has a statistical cost; exact universal improvement over the target-only baseline should not be promised without conditions. Your covariate-shift work makes this distinction natural: correct specification and misspecification can call for different procedures.^{5,6}

The first empirical test can hold pretrained embeddings fixed, avoiding a large pretraining effort. Use controlled changes in correlation, noise, missingness, and target dependence to test the predicted regimes. Then test one scientifically meaningful setting, such as a multimodal measurement problem with a collaborator-defined target. Evaluate downstream prediction and uncertainty, including rare groups or acquisition conditions when relevant; reconstruction and representation alignment alone do not answer the proposed question.

The main risk is an abstraction that omits the useful phenomenon. Linear Gaussian models are valuable for isolating leakage or identifiability, but their conclusions do not automatically describe nonlinear semantic representations. The contrastive paper itself illustrates limitations when a representation-class condition fails. Use the simple model to make a falsifiable prediction, then test whether that prediction survives approximate shared structure and nonlinear features.¹⁰

Continue if the theory predicts a practical choice of modality, regularization, or labeling strategy that standard baselines miss. If the final result only ranks subspace errors without changing a downstream choice, keep it as focused methodological work rather than making it the central AI-facing claim.

10. Gui, Ma and Ma, *Multi-Modal Contrastive Learning Adapts...*

11. Gui, Ma and Ma, *IndiSeek Learns Information-Guided...*

12. Yang and Ma, *Estimating Shared Subspace with AJIVE:...*

13. Song, Yang and Ma, *Bias-Corrected Subspace Intersection:...*

16. Yang and Ma, *Sample-Efficient Inductive Matrix...*

36. Dufumier et al., *What to Align in Multimodal Contrastive...*

5. Ma et al., *Optimally Tackling Covariate Shift in...*

6. Ge et al., *Maximum Likelihood Estimation is All You...*

Two connecting investments: adaptive computation and synthetic evidence

Adaptive computation is a natural use of the bandit work. The question is whether the next unit of effort should generate another candidate, extend reasoning, call a different verifier, retrieve external evidence, or obtain an expert check. The objective should measure useful validated output against total cost, including latency and human time. This is more informative than optimizing token count in isolation.

Your batched-bandit work proves a price for adapting to an unknown margin under limited updates in its specified nonparametric model. It does not automatically imply a lower bound for reasoning systems, but it suggests a productive question: when is parallel generation an adequate substitute for another round of feedback, and when does limited adaptivity create an unavoidable loss?¹⁹

This territory already has substantial work. Snell and colleagues show that prompt difficulty affects useful compute allocation. A 2026 selective-verification study finds that a longer initial solve can match or beat more elaborate verification policies at lower total cost in its tested settings. These are reasons to demand strong baselines, including a simple longer solve, uniform sampling, and best-of-N with the same verifier, before investing in a new controller.^{37,38}

A bounded project would start with two available interventions and a known trusted-check mechanism. Learn the allocation policy from logged interventions, then add batching, unknown problem difficulty, or correlated verifier errors one at a time. The interesting contribution is a decision rule with a meaningful guarantee under uncertainty about intervention value. Keep this within the evaluation or post-training project until the mechanism warrants an independent line.

Synthetic data belongs inside the feedback question. Generating more samples can expose useful knowledge already encoded in a model, reduce optimization difficulty, or exploit new external feedback. It does not automatically provide independent evidence about a target population. Count the origin and dependence of information explicitly: fresh measurements, a pretrained model's prior information, a verifier, and repeated samples from the same generator play different roles.

The literature also rules out a simple prediction that synthetic data must cause collapse. Accumulating original and synthetic data can avoid the deterioration seen in replacement-based recursions in studied settings. Meanwhile, finite-sample PPI analyses show that estimating how much to trust a predictor can itself be costly; filtered PPI already addresses heterogeneous prediction quality. A new project must specify the feedback and shift mechanism left unresolved by these results.^{28,39,40}

A useful longer-term question is the minimum amount and placement of fresh, trusted evidence required to sustain reliable improvement over many learning rounds. Start with a fixed target and controlled error structure, rather than a theorem about arbitrary self-improving systems. This could eventually unify the program: demonstrations supply candidate behavior, multimodal data supply representations, and adaptive audits determine whether the apparent improvement is real.

19. Jiang and Ma, *The Adaptivity Barrier in Batched...*

37. Snell et al., *Scaling LLM Test-Time Compute Optimally...*

38. Dip, Zhou and Zhang, *Think Again or Think Longer? Selective...*

28. Mani et al., *No Free Lunch: Non-Asymptotic Analysis...*

39. Gerstgrasser et al., *Is Model Collapse Inevitable? Breaking...*

40. Xu and Sun, *Do More Predictions Improve Statistical...*

Use AI to increase research depth and train stronger researchers

Make AI assistance routine, but evaluate it through verified research progress. For each project, keep a short record of the scientific question, the current conjecture, the assumptions that matter, the strongest baseline, and the evidence that would change the conclusion. Use AI to search for counterexamples, implement diagnostic experiments, explore alternative decompositions, and propose proof steps. Judge the output by what survives checking and improves understanding.

Formalization is a useful continuation of your existing BCSI practice. A bounded research investment could extract reusable components for matrix perturbation, probability inequalities, and statistical lower bounds from actual papers. Prefer components that a second project immediately needs. A full general-purpose automated theorem prover would require a different research program; no evidence in the reviewed portfolio makes that the best primary investment.

Distinguish three checks. First, the proof must establish its formal statement. Second, the statement and definitions must match the intended mathematics. Third, the assumptions and target must support the scientific interpretation. A successful Lean build addresses the first check; the remaining checks require separate scrutiny. The public BCSI repository reports theorem-entry verification, which is a promising basis for this practice, not a reason to skip the other checks.¹⁴

For students, use two complementary modes. In learning mode, require them to reconstruct central arguments, explain why hypotheses are necessary, and solve small variants without assistance. In research mode, permit aggressive use of AI while requiring ownership of definitions, proof dependencies, counterexamples, and experimental interpretation. Being able to supervise a powerful tool depends on having enough unaided understanding to recognize when it is wrong.

A useful group meeting would compare the current explanation with its strongest counterexample and simplest baseline. Ask what remains unexplained, whether the same result follows from existing machinery, and what practical conclusion would change if the conjecture were true. The aim is to prevent faster execution from committing the group to weak questions. Reward a well-supported decision to stop a project as well as a successful proof.

Track outcomes such as a new mechanism isolated, an incorrect conjecture eliminated, a baseline overturned under reproducible conditions, or a reusable theorem independently applied. Numbers of prompts, generated lemmas, or pages do not measure research value. A periodic internal assessment on held-out tasks from your own work would also be more relevant to staffing and workflow choices than headline benchmark scores.

Preserve intellectual depth. There is no need to attach an AI application to every important theorem. Maintain space for classical statistical questions with substantial consequences, and for mathematical skills that make unfamiliar arguments intelligible. The strategic change is to demand a clearer account of why a result matters while using improving tools to pursue harder versions of that result.

14. Cong Ma et al., *BCSI Lean Verification*

A practical research portfolio for the next twelve months

Start two projects with one shared experimental setting. The most coherent pairing is adaptive evaluation and learning with imperfect verification. A common environment supplies prompts, generated answers, a changing evaluator, and trusted outcome checks. The first project asks whether improvement can be measured reliably; the second asks how to obtain it. This permits distinct theoretical questions while sharing the effort of building credible evidence.

A reasonable allocation of discretionary research attention is 40% to reliable evaluation, 30% to demonstrations and verification, 20% to a focused multimodal continuation, and 10% to formalization and exploratory work. These are planning weights, not estimated returns or a prescription for every student. Existing commitments, student interests, funding, and collaborator access should determine the actual allocation.

Months 1-3: write one precise problem statement for each flagship, map it against the nearest papers, and reproduce the relevant simple baselines. Construct one example where the current method fails for the proposed reason. Establish what is observed, what is adaptively chosen, what is trusted, and what target remains fixed. Run a small empirical pilot before committing to a long proof.

Months 4-6: prove a theorem in the smallest model that retains the failure mechanism, and seek a lower bound or counterexample showing why its key condition is needed. In parallel, test the predicted behavior with frozen models and limited interventions. This stage should resolve whether the project has a conceptual contribution or merely a technically elaborate combination of known components.

Months 7-9: challenge the result under one realistic departure from the initial assumptions. For evaluation, this could be a shared bias across judges or a model update chosen from previous audit outcomes. For learning, it could be verifier errors concentrated on newly generated answers. For multimodal transfer, it could be a useful signal present only in one view. Expand only the project whose prediction survives.

Months 10-12: deliver a paper with a clear statistical principle, a reproducible comparison, and explicit limits. Formalize the central reusable argument where that is useful. Release an evaluation protocol or compact implementation that another group can use. Choose the next project based on the limitations revealed, rather than on a predetermined list of extensions.

Before launching any additional line, apply four tests: does it address a consequential decision; does it expose a mechanism not handled by the strongest existing result; can the group obtain credible evidence at feasible cost; and will the contribution remain meaningful if models become much more capable? A project need not pass every test equally, but a difficult proof with weak answers to all four should receive low priority.

The intended research identity is precise: statistical principles for learning, evaluating, and deciding with imperfect information supplied by AI and other data sources. Your existing work provides several routes into that program. The strongest next move is to connect them through a few questions whose importance grows as AI becomes more capable.

Sources

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2. **Ma, Wang, Chi and Chen.** *Implicit Regularization in Nonconvex Statistical Estimation*. 2017; FOCM 2020.
3. **Chen, Fan, Ma and Yan.** *Inference and Uncertainty Quantification for Noisy Matrix Completion*. PNAS, 2019.
4. **Tong, Ma and Chi.** *Accelerating Ill-Conditioned Low-Rank Matrix Estimation via Scaled Gradient Descent*. 2020; JMLR 2021.
5. **Ma, Pathak and Wainwright.** *Optimally Tackling Covariate Shift in RKHS-Based Nonparametric Regression*. 2022; Annals of Statistics 2023.
6. **Ge, Tang, Fan, Ma and Jin.** *Maximum Likelihood Estimation is All You Need for Well-Specified Covariate Shift*. 2023; ICLR 2024.
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8. **Lee and Ma.** *Off-Policy Estimation with Adaptively Collected Data: The Power of Online Learning*. NeurIPS, 2024.
9. **Joshi et al..** *Learning to Answer from Correct Demonstrations*. 2025; v2 February 2026; ICLR 2026. See Sections 6 and 8 for pass@k and computational limitations.
10. **Gui, Ma and Ma.** *Multi-Modal Contrastive Learning Adapts to Intrinsic Dimensions of Shared Latent Variables*. NeurIPS, 2025. Appendix G.2 illustrates a failure of semantic alignment when a representation-class condition fails.
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